

## 1. Retrieve Kunitz-Domain Proteins

*QUERY: ( Identifier = "PF00014" AND Annotation Type = "Pfam" ) AND Polymer Entity Sequence Length >= 40 AND Polymer Entity Sequence Length <= 80 AND Refinement Resolution <= 3.5*

 Output: `rscsb_pdb_custom_report_20250606082417.csv` ⇒ Downloaded protein metadata containing Kunitz domain from RCSB PDB.

## 2. Extract Sequences to FASTA Format

*cat rscsb\_pdb\_custom\_report\_20250606082417.csv | tr -d '"' | awk -F ';' '{if (length(\$2) > 0) {name = \$2}; print name, \$3, \$4, \$5}' | grep PF00014 | awk '{print ">"\$1"\_"\$3; print \$2}' > pdb\_kunitz.fasta*

 Output: `pdb_kunitz.fasta` ⇒ Extracted sequence data from the metadata file into FASTA format.

## 3. Cluster Sequences Using CD-HIT

*cd-hit -i pdb\_kunitz.fasta -o pdb\_kunitz.clust*

 Outputs:

- `pdb_kunitz.clust` ← Non-redundant sequences
- `pdb_kunitz.clust.clstr` ← Clustering summary ⇒ Removed highly similar sequences to ensure diversity in model training.

## 4. Rename Final FASTA for Model Input

*mv pdb\_kunitz.clust pdb\_kunitz\_nr.fasta*

 Output: `pdb_kunitz_nr.fasta` ⇒ Renamed clustered FASTA file for consistency and clarity.

## 5. Filter and Format PDB IDs

*grep ">" pdb\_kunitz\_nr.fasta | grep -v 2ODY\_E | tr -d ">" | tr " " ":" > pdb\_list.txt*

 Output: `pdb_list.txt`    ⇨ Extracted PDB identifiers for downstream structure-based alignment.

## 6a. PDBeFold (Structure-Based Alignment)

 Input: `pdb_list.txt`

 Outputs:

- `fasta.seq, match.dat`
- Renamed as `pdb_kunitz_nr.ali`    ⇨ Aligned sequences structurally for high-confidence model input.

## 6b. MUSCLE (Sequence-Based Alignment)

 Input: `pdb_kunitz_nr.fasta`

 Output: `*.sequence`    ⇨ Sequence-based multiple alignment for comparative modeling.

## 7. Format Alignment for HMMER Input

```
awk '{if (substr($1,1,1)==">") {print "\n"toupper($1)} else {printf "%s", toupper($1)}}' \
    pdb_kunitz_nr.ali | sed 's/PDB:/' | tail -n +2 > pdb_kunitz_nr_clean.ali
```

 Output: `pdb_kunitz_nr_clean.ali`    ⇨ Re-encoded alignment into format compatible with HMMER.

## 8. Build Hidden Markov Model (HMM)

```
hmmbuild pdb_kunitz_nr_clean.hmm pdb_kunitz_nr_clean.ali
```

 Output: `pdb_kunitz_nr_clean.hmm`    ⇨ Built the statistical model using `hmmbuild`.

## 9. Preparing Data for Testing the Model

*QUERY: (xref:pfam-pf00014 and reviewed: true)*

```

QUERY: ( reviewed: true)
grep ">" ../../uniprot_sprot.fasta |cut -d "|" -f2 |sort > sp.ids
grep ">" ../../all_kunitz.fasta |cut -d "|" -f2 |sort > all_kunitz.ids
comm -23 ../id_files/sp.ids ../id_files/all_kunitz.ids > sp_negs.ids
python get_seq.py sp_negs.id ../../swissprot_data/uniprot_sprot.fasta > sp_negs.fasta

```

 Outputs:

- `all_kunitz.fasta` ← 398 positive proteins
- `uniprot_sprot.fasta` ← 573,230 reviewed Swiss-Prot proteins
- `sp.ids`, `all_kunitz.ids`, `sp_negs.ids`
- `sp_negs.fasta` ← 572,832 negative protein sequences    ⇨ Collected and separated test data for performance evaluation.

## 10. Filtering Test Set to Remove Redundant Matches with Training

```

makeblastdb -in ../../swissprot_data/all_kunitz.fasta -input_type fasta -dbtype prot -out
all_kunitz.fasta

```

```

blastp -query ../PDB_data/pdb_kunitz_nr.fasta -db ../../all_kunitz.fasta -out
../../pdb_kunitz_nr_23.blast -outfmt 7

```

```

grep -v "^#" pdb_kunitz_nr_23.blast |awk '{if ($3 >=95 && $4>=50) print $2}' |cut -d "|" -f 2 >
../id_files/to_remove_ids
comm -23 <(sort all_kunitz.ids) <(sort to_remove_ids) >ok_kunitz.ids

```

```

python ../../get_seq.py ok_kunitz.ids ../../all_kunitz.fasta > ../../ok_kunitz.fasta

```

 Inputs: `pdb_kunitz_nr.fasta`, `all_kunitz.fasta`

 Outputs:

- `all_kunitz.fasta.pdb` ← BLAST database
- `pdb_kunitz_nr_23.blast` ← BLAST results
- `to_remove_ids` ← 32 overlapping sequences
- `ok_kunitz.ids` ← 367 filtered positives
- `ok_kunitz.fasta` ← final test-positive dataset    ⇨ Removed test sequences with high similarity to training data.

## 11. Constructing Balanced Positive and Negative Test Sets

```
head -n 185 random_ok_kunitz.ids > pos_1.ids
tail -n 185 random_ok_kunitz.ids > pos_2.ids
head -n 286416 random_sp_negs.id > neg_1.ids
tail -n +286417 random_sp_negs.id > neg_2.ids
```

```
python get_seq.py ../pos_1.ids ../swissprot_data/uniprot_sprot.fasta > pos_1.fasta
grep ">" ../final_files/pos_1.fasta |wc ⇒ 184
```

```
python get_seq.py ../s/pos_2.ids ../swissprot_data/uniprot_sprot.fasta > pos_2.fasta
grep ">" ../final_files/pos_2.fasta |wc ⇒ 184
```

```
python get_seq.py ../seq_ids/neg_1.ids ../swissprot_data/uniprot_sprot.fasta > neg_1.fasta
grep ">" ../final_files/neg_1.fasta |wc ⇒ 286416
```

```
python get_seq.py ../seq_ids/neg_2.ids ../swissprot_data/uniprot_sprot.fasta > neg_2.fasta
grep ">" ../final_files/neg_2.fasta |wc ⇒ 286416
```

 Inputs:

- `ok_kunitz.ids` ← 367 filtered positives
- `sp_negs.ids` ← 572,832 negatives

 Outputs:

- `pos_1.fasta`, `pos_2.fasta` ← each with 184 sequences
- `neg_1.fasta`, `neg_2.fasta` ← each with 286,416 sequences ⇒ Randomly partitioned test datasets for balanced evaluation.

## 12. Running HMMsearch on Test Sets and Generating `.class` Files

```
hmmsearch -Z 1000 --max --tblout pos_1.out ../HMM/pdb_kunitz_nr_clean.hmm pos_1.fasta
grep -v "^#" pos_1.out |awk '{split($1, a, "|") ; print a[2],1,$5,$8}' |tr " " "\t" > pos_1.class
```

```
hmmsearch -Z 1000 --max --tblout pos_2.out ../HMM/pdb_kunitz_nr_clean.hmm pos_2.fasta
grep -v "^#" pos_2.out |awk '{split($1, a, "|") ; print a[2],1,$5,$8}' |tr " " "\t" > pos_2.class
```

```
hmmsearch -Z 1000 --max --tblout neg_1.out ../HMM/pdb_kunitz_nr_clean.hmm neg_1.fasta
grep -v "^#" neg_1.out |awk '{split($1, a, "|") ; print a[2],1,$5,$8}' |tr " " "\t" > neg_1.class
```

```
hmmsearch -Z 1000 --max --tblout neg_2.out ../HMM/pdb_kunitz_nr_clean.hmm neg_2.fasta
grep -v "^#" neg_2.out |awk '{split($1, a, "|") ; print a[2],1,$5,$8}' |tr " " "\t" > neg_2.class
```

```
cat pos_1.class neg_1.class > set_1.class
cat pos_2.class neg_2.class > set_2.class
```

## Inputs:

- pos\_1.fasta, pos\_2.fasta
- neg\_1.fasta, neg\_2.fasta
- pdb\_kunitz\_nr\_clean.hmm

## Outputs:

- HMMsearch results: pos\_1.out, pos\_2.out, neg\_1.out, neg\_2.out
- Class files: pos\_1.class, pos\_2.class, neg\_1.class, neg\_2.class
- Merged datasets: set\_1.class, set\_2.class ⇒ Executed HMM-based similarity search and formatted results for ML modeling.

## 13. Evaluating HMM Model Performance Using Multiple E-Value Thresholds

for i in 'seq 1 12'; do python [performance.py](#) set\_1.class 1e-\$i; done  
for i in 'seq 1 12'; do python [performance.py](#) set\_2.class 1e-\$i; done

## Inputs: set\_1.class, set\_2.class, performance.py

## Outputs:

- Q2, MCC, TPR, PPV metrics across 12 thresholds (1e-1 to 1e-12) ⇒ Quantified model performance for discussion and optimization.

```
(base) sarehsalehi@DESKTOP-FAB9J57:/mnt/d/bioinfo-semester2/Laboratory-of-bioinformatics-1/Module-Emidio/project/final_test_2$ for i in $(seq 1 12); do python t_1.class 1e-$i; done
TH=0.1      Q2=0.9998813677599442  MCC=0.9186601672859828  TPR=0.8440366972477065  PPV=1.0      CM=[[286382, 0], [34, 184]]
TH=0.01     Q2=0.999940683879972  MCC=0.9567489489700443  TPR=0.9154228855721394  PPV=1.0      CM=[[286399, 0], [17, 184]]
TH=0.001    Q2=0.9999860432658758  MCC=0.9892975971352793  TPR=0.9787234042553191  PPV=1.0      CM=[[286412, 0], [4, 184]]
TH=0.0001   Q2=0.9999930216329379  MCC=0.9946056525388577  TPR=0.989247311827957  PPV=1.0      CM=[[286414, 0], [2, 184]]
TH=1e-05    Q2=0.9999965108164689  MCC=0.9972918941022746  TPR=0.9945945945945946  PPV=1.0      CM=[[286415, 0], [1, 184]]
TH=1e-06    Q2=1.0      MCC=1.0      TPR=1.0      PPV=1.0      CM=[[286416, 0], [0, 184]]
TH=1e-07    Q2=0.9999965108164689  MCC=0.9972771655630667  TPR=1.0      PPV=0.9945652173913043  CM=[[286416, 1], [0, 183]]
TH=1e-08    Q2=0.9999965108164689  MCC=0.9972771655630667  TPR=1.0      PPV=0.9945652173913043  CM=[[286416, 1], [0, 183]]
TH=1e-09    Q2=0.9999965108164689  MCC=0.9972771655630667  TPR=1.0      PPV=0.9945652173913043  CM=[[286416, 1], [0, 183]]
TH=1e-10    Q2=0.9999965108164689  MCC=0.9972771655630667  TPR=1.0      PPV=0.9945652173913043  CM=[[286416, 1], [0, 183]]
TH=1e-11    Q2=0.9999965108164689  MCC=0.9972771655630667  TPR=1.0      PPV=0.9945652173913043  CM=[[286416, 1], [0, 183]]
TH=1e-12    Q2=0.99998953244494068  MCC=0.9918091292092845  TPR=1.0      PPV=0.9836956521739131  CM=[[286416, 3], [0, 181]]
(base) sarehsalehi@DESKTOP-FAB9J57:/mnt/d/bioinfo-semester2/Laboratory-of-bioinformatics-1/Module-Emidio/project/final_test_2$ for i in $(seq 1 12); do python t_2.class 1e-$i; done
TH=0.1      Q2=0.9998639218422889  MCC=0.9082949717777876  TPR=0.8251121076233184  PPV=1.0      CM=[[286377, 0], [39, 184]]
TH=0.01     Q2=0.9999616189811584  MCC=0.9713668227750496  TPR=0.9435897435897436  PPV=1.0      CM=[[286405, 0], [11, 184]]
TH=0.001    Q2=0.9999825510823447  MCC=0.98644447630067864  TPR=0.9837837837837838  PPV=0.9891304347826086  CM=[[286413, 2], [3, 182]]
TH=0.0001   Q2=0.999971575757152826  MCC=0.9809178567341356  TPR=0.9836065573770492  PPV=0.9782608695652174  CM=[[286413, 4], [3, 180]]
TH=1e-05    Q2=0.9999860432658758  MCC=0.9890638036334939  TPR=1.0      PPV=0.9782608695652174  CM=[[286416, 4], [0, 180]]
TH=1e-06    Q2=0.9999860432658758  MCC=0.9890638036334939  TPR=1.0      PPV=0.9782608695652174  CM=[[286416, 4], [0, 180]]
TH=1e-07    Q2=0.9999825540823447  MCC=0.9863108559202272  TPR=1.0      PPV=0.9728260869565217  CM=[[286416, 5], [0, 179]]
TH=1e-08    Q2=0.9999825540823447  MCC=0.9863108559202272  TPR=1.0      PPV=0.9728260869565217  CM=[[286416, 5], [0, 179]]
TH=1e-09    Q2=0.9999825540823447  MCC=0.9863108559202272  TPR=1.0      PPV=0.9728260869565217  CM=[[286416, 5], [0, 179]]
TH=1e-10    Q2=0.9999685973482205  MCC=0.9752215500189169  TPR=1.0      PPV=0.9510869565217391  CM=[[286416, 9], [0, 175]]
TH=1e-11    Q2=0.9999685973482205  MCC=0.9752215500189169  TPR=1.0      PPV=0.9510869565217391  CM=[[286416, 9], [0, 175]]
TH=1e-12    Q2=0.9999616189811584  MCC=0.9696294565487017  TPR=1.0      PPV=0.9402173913043478  CM=[[286416, 11], [0, 173]]
(base) sarehsalehi@DESKTOP-FAB9J57:/mnt/d/bioinfo-semester2/Laboratory-of-bioinformatics-1/Module-Emidio/project/final_test_2$
```

